AP CALCULUS AB

Unit 1 • Lesson 1.11 practice

Lesson 1.11 • Defining Continuity at a Point

5 multiple-choice questions and 1 free-response question

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Let $f(x) = x^2$ for $x \neq 2$ and $f(2) = 1$. Which condition for continuity at $x = 2$ fails?

(A) $f(2)$ is defined.
(B) $\lim_{x \to 2} f(x)$ exists.
(C) $\lim_{x \to 2} f(x) = f(2)$.
(D) None; f is continuous at 2.

Question 2

NO CALCULATOR

Let $f(x) = \begin{cases} 2x + 1, & x < 1 \\ 3, & x = 1 \\ x^2 + 2, & x > 1. \end{cases}$ Is f continuous at $x = 1$?

(A) Yes
(B) No, $f(1)$ is undefined.
(C) No, the limit does not exist.
(D) No, the limit is not equal to $f(1)$.

Question 3

NO CALCULATOR

Why is $f(x) = \dfrac{x^2 - 9}{x - 3}$ not continuous at $x = 3$?

(A) The limit does not exist.
(B) $f(3)$ is undefined.
(C) The limit is infinite.
(D) It is continuous at $x = 3$.

Question 4

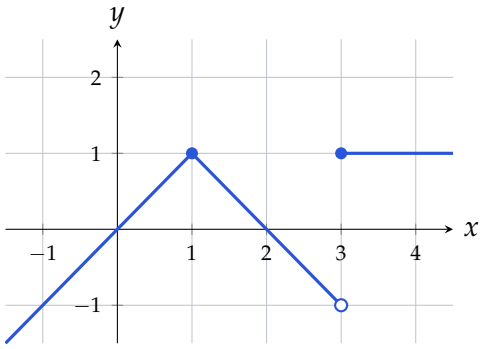
NO CALCULATOR

Which function is continuous at $x = 0$?

(A) $\dfrac{1}{x}$
(B) $\dfrac{|x|}{x}$
(C) $\dfrac{\sin x}{x}$
(D) $|x|$

Question 5 (graph) NO CALCULATOR

The graph of f is shown. Which statement is true?



(A) f is continuous at $x = 3$ but not at $x = 1$.
(B) f is continuous at $x = 1$ but not at $x = 3$.
(C) f is continuous at both $x = 1$ and $x = 3$.
(D) f is continuous at neither $x = 1$ nor $x = 3$.

Question 6 (Free response) NO CALCULATOR

Let $f(x) = \begin{cases} x^2 - 1, & x < 2 \\ 3, & x = 2 \\ 5 - x, & x > 2. \end{cases}$

(a) Find $\lim_{x \to 2} f(x)$, showing both one-sided limits.
(b) Is f continuous at $x = 2$? Justify using the definition of continuity.
(c) Is f continuous at $x = 0$? Explain.